Aircraft Performance Engine

Takeoff & Landing Performance Model

Methodology Report: Formulas, Methods and Worked Examples

Atmosphere, aircraft, takeoff, Balanced Field Length, landing and runway analysis,
organised by Python file and function, with worked examples for the
Bombardier Challenger 3500 and the Cirrus SR22T

Mark Barsoum

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Summary

This report sets out the methodology behind a Python model of aircraft takeoff and landing performance. It does not evaluate the results; it explains, step by step, how they are obtained.

Part I follows the structure of the code. Each section covers one Python file, and each subsection one function: what it takes in, what it returns, the formulas it evaluates and the source of each method. The files build on one another, from unit conversion and the atmosphere, through the aircraft model and the equations of motion, to the takeoff distance, the Balanced Field Length (BFL) with an engine failure at the decision speed V_1, the landing distance and the runway checks.

Part II applies the method to two aircraft on the same runway and day: a Bombardier Challenger 3500 twin-engine business jet and a Cirrus SR22T single-engine turbocharged piston aircraft. Every intermediate value is shown with the numbers substituted into the equations of Part I. The worked examples are generated directly from the model by `report/make_examples.py`, so the numbers in the report always match the code.

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1 Introduction

1.1 Purpose

Takeoff and landing performance determines which runways an aircraft can use and at what weight. For transport-category aircraft, the certified takeoff distance is set by the most critical engine-failure case, not only by the distance with all engines running. This report documents a model that computes these distances from first principles and applies the certification rules of 14 CFR Part 25 and Transport Canada CAR 525.

1.2 How the Code Is Organised

Table 1 lists the Python files in the order they build on each other. Part I of this report follows the same order.

Section	File	Role
2	`units.py`	Unit conversion functions
3	`atmosphere.py`	International Standard Atmosphere (ISA)
4	`aircraft.py`	Aircraft classes: aerodynamics and thrust
5	`segments.py`	Day conditions and the equations of motion
6	`takeoff.py`	All-engines takeoff distance
7	`bfl.py`	Balanced Field Length and Takeoff Field Length
8	`landing.py`	Landing distance and Landing Distance Required
9	`airports.py`	Runways, declared distances and wind components
10	`runway_analysis.py`	Runway checks with regulatory wind factors
11	`sensitivity.py`	Parameter sweeps
12	others	Exceptions, profiles, charts and the menu

Table 1. Python files and their role.

1.3 Conventions

- All calculations are in SI units. Feet, knots, pounds and degrees Celsius are used only for input and output.
- **All speeds are true airspeed (TAS)**, labelled KTAS when shown in knots.
- Sources are cited by number in IEEE style, e.g. [1]; full citations are in the References, numbered in order of first citation.

1.4 Nomenclature

Symbol	Meaning	Unit
a	Speed of sound	m/s
C_L, C_D	Lift and drag coefficients	–
C_{D_0}	Zero-lift (parasite) drag coefficient	–
$C_{L,\text{gnd}}$	Lift coefficient during the ground roll	–
$C_{L,\max}$	Maximum lift coefficient (takeoff or landing flaps)	–
D, L, T, W	Drag, lift, thrust, weight	N
g_0	Standard gravity, 9.80665	m/s^2
h	Pressure altitude / field elevation	m
h_{scr}	Screen height	m
h_{crit}	Turbocharger critical altitude	m
k	Induced-drag factor	–
k_{lapse}	Turbofan thrust lapse with Mach number	–
L_{r}	Temperature lapse rate, 0.0065	K/m
M	Mach number	–
m	Aircraft mass	kg
n	Load factor in the transition arc	–
N_e, N_{op}	Number of engines; engines operating	–
P	Shaft power	W
p	Static pressure	Pa
q	Dynamic pressure, $\frac{1}{2}\rho V^2$	Pa
R	Specific gas constant of air, 287.05	J/(kg K)
R_{TR}	Transition arc radius	m
S	Reference wing area	m^2
s	Distance along the runway	m
T_{SLS}	Sea-level static thrust, all engines	N
Θ	Static air temperature	K
ΔT	ISA deviation	°C
ΔT_{flat}	Flat-rating temperature	°C
V, V_{TAS}	True airspeed	m/s
V_{GS}	Ground speed	m/s
V_w, V_x	Headwind (negative = tailwind) and crosswind components	m/s
V_{S1}, V_{S0}	Stall speed, takeoff and landing configuration	m/s

Nomenclature (continued)

Symbol	Meaning	Unit
V_1, V_R, V_2	Decision, rotation and takeoff safety speeds	m/s
V_{REF}, V_{TD}	Landing reference and touchdown speeds	m/s
γ	Climb angle	rad
γ_{air}	Ratio of specific heats, 1.4	–
η_p	Propeller efficiency	–
θ	Runway slope angle (+ uphill)	rad
μ_{roll}, μ_{brake}	Rolling and braking friction coefficients	–
ρ	Air density	kg/m^3
σ	Density ratio, ρ/ρ_{SL}	–

Table 2. Nomenclature.

PART I

Methodology by Python File

2 `units.py`: Unit Conversion

Every conversion in the project goes through this file, so each factor is defined exactly once [2]. Each function accepts a single number or a NumPy array.

Functions	Conversion	Factor
`ft_to_m()` / `m_to_ft()`	feet ↔ metres	1 ft = 0.3048 m (exact)
`ft2_to_m2()` / `m2_to_ft2()`	$\text{ft}^2 \leftrightarrow \text{m}^2$	0.3048^2
`kt_to_mps()` / `mps_to_kt()`	knots ↔ m/s	1 kt = 1852/3600 m/s (exact)
`lb_to_kg()` / `kg_to_lb()`	pound-mass ↔ kg	1 lb = 0.45359237 kg (exact)
`lbf_to_n()` / `n_to_lbf()`	pound-force ↔ N	1 lbf = 4.4482216 N
`hp_to_w()` / `w_to_hp()`	horsepower ↔ W	1 hp = 745.69987 W
`c_to_k()` / `k_to_c()`	°C ↔ K	$+273.15$

Table 3. Conversion functions in `units.py`.

3 `atmosphere.py`: International Standard Atmosphere

Method: [3, Ch. 3]; constants from [4]. Valid in the troposphere, $0 \le h \le 11\,000$ m, with sea-level values $\Theta_{\text{SL}} = 288.15$ K, $p_{\text{SL}} = 101\,325$ Pa and $\rho_{\text{SL}} = 1.225$ kg/m^3.

3.1 `isa()`

`isa(alt_m, isa_dev_c=0)` → `(temp_k, press_pa, rho_kg_m3)`
Returns: temperature, pressure and density at a pressure altitude.

The standard temperature falls linearly with altitude. Pressure follows the standard profile, so an ISA deviation changes temperature and density but not pressure:

$$\Theta_{\text{std}} = \Theta_{\text{SL}} - L_{\text{r}}\, h \tag{3.1}$$

$$p = p_{\text{SL}} \left(\frac{\Theta_{\text{std}}}{\Theta_{\text{SL}}} \right)^{g_0/(L_{\text{r}} R)}, \qquad \frac{g_0}{L_{\text{r}} R} \approx 5.2559 \tag{3.2}$$

$$\Theta = \Theta_{\text{std}} + \Delta T \tag{3.3}$$

$$\rho = \frac{p}{R\,\Theta} \tag{3.4}$$

An altitude outside the valid range raises `AtmosphereRangeError`.

3.2 `sigma()`

`sigma(alt_m, isa_dev_c=0)` → `float`
Returns: the density ratio σ.

$$\sigma = \frac{\rho}{\rho_{\text{SL}}} \tag{3.5}$$

3.3 `speed_of_sound()`

`speed_of_sound(alt_m, isa_dev_c=0)` → `float`
Returns: the speed of sound [m/s].

$$a = \sqrt{\gamma_{\text{air}}\, R\, \Theta} \tag{3.6}$$

4 `aircraft.py`: Aircraft Model

An `Aircraft` base class holds the geometry, aerodynamics and friction common to every aircraft. Two subclasses override the behaviour that differs by engine type:

- `PropAircraft`: propeller thrust (normally aspirated or turbocharged), 50 ft takeoff screen height [5], 1 s rotation, 2 s free roll.
- `JetAircraft`: flat-rated turbofan thrust, ground spoilers and reverse thrust, 35 ft takeoff screen height [6], [7], 3 s rotation, 3 s free roll, LDR rule.

4.1 `Aircraft.lift()` and `Aircraft.drag()`

`lift(tas_mps, rho_kg_m3, cl)`, `drag(tas_mps, rho_kg_m3, cl)` → `float`
Returns: lift and drag [N].

A parabolic drag polar [8, Ch. 6]:

$$q = \tfrac{1}{2}\rho V^2 \tag{4.1}$$

$$L = q\, S\, C_L \tag{4.2}$$

$$D = q\, S \left(C_{D_0} + k\, C_L^2\right) \tag{4.3}$$

4.2 `Aircraft.stall_speed()`

`stall_speed(weight_n, rho_kg_m3, cl_max)` → `float`
Returns: stall speed, TAS [m/s].

$$V_S = \sqrt{\frac{2W}{\rho\, S\, C_{L,\max}}} \tag{4.4}$$

V_{S1} uses $C_{L,\max}$ with takeoff flaps; V_{S0} uses landing flaps.

4.3 `Aircraft.engine_fraction()`

`engine_fraction(engines_operating)` → `float`
Returns: the share of total thrust available.

$$T = T_{\text{AEO}}\,\frac{N_{\text{op}}}{N_e}, \qquad N_{\text{op}} = N_e - 1 \text{ with one engine inoperative (OEI)} \tag{4.5}$$

4.4 `PropAircraft.thrust()`

`thrust(tas_mps, alt_m, isa_dev_c, engines_operating)` → `float`
Returns: propeller thrust [N].

A constant-efficiency propeller gives $T = \eta_p P/V$, which goes to infinity as $V \to 0$, so it is capped at the static thrust [9]; [8]. A normally aspirated piston engine loses power with density; a turbocharged engine holds rated power up to its critical altitude:

$$\sigma_P = \begin{cases} \sigma(h, \Delta T), & \text{normally aspirated} \\ 1, & \text{turbocharged},\ h \le h_{\text{crit}} \\ \sigma(h, \Delta T)/\sigma(h_{\text{crit}}, \Delta T), & \text{turbocharged},\ h > h_{\text{crit}} \end{cases} \tag{4.6}$$

$$T_{\text{prop}} = \min\left(\sigma_P\, T_{\text{SLS}},\ \frac{\eta_p\, \sigma_P\, P_{\max}}{\max(V,\, 1)}\right)\frac{N_{\text{op}}}{N_e} \tag{4.7}$$

4.5 `JetAircraft.thrust()`

`thrust(tas_mps, alt_m, isa_dev_c, engines_operating)` → `float`
Returns: turbofan thrust [N].

The engine control holds rated thrust on warm days up to the flat-rating temperature. Only the temperature above it reduces thrust (simplified from [10]):

$$\Delta T_{\text{excess}} = \max\left(\Delta T - \Delta T_{\text{flat}},\ 0\right) \tag{4.8}$$

$$\sigma_{\text{eff}} = \sigma(h,\ \Delta T_{\text{excess}}) \tag{4.9}$$

$$T_{\text{jet}} = T_{\text{SLS}}\,\sigma_{\text{eff}}^{0.7}\left(1 - k_{\text{lapse}}\,\frac{V}{a(h,\Delta T)}\right)\frac{N_{\text{op}}}{N_e} \tag{4.10}$$

4.6 `cl_braking()` and `reverse_thrust()`

`cl_braking()` → `float`; `reverse_thrust(tas_mps, use_reverse)` → `float`
Returns: lift coefficient while braking; reverse thrust [N].

On a jet, ground spoilers dump the lift, so $C_L = 0$ while braking; a propeller aircraft keeps $C_{L,\text{gnd}}$. Reverse thrust is [11]

$$T_{\text{rev}} = \begin{cases} r_{\text{rev}}\,T_{\text{SLS}}, & V > 60\,\text{kt} \\ 0, & \text{otherwise} \end{cases} \tag{4.11}$$

It is not credited in certified dry-runway distances [6, §25.125], so it is switched off in the accelerate-stop case and in certified landing distances.

5 `segments.py`: Conditions and Equations of Motion

The building blocks every takeoff and landing calculation is made of. The equations of motion are written once here and reused by `takeoff.py`, `bfl.py` and `landing.py`.

5.1 `make_conditions()`

`make_conditions(acft, mass_kg, elev_m, isa_dev_c, hw_kt, slope_pct, rwy_condition)` → `Conditions`
Returns: the day's density, weight, wind, slope and braking friction.

Weight is $W = m\,g_0$ and density comes from `isa()`. The runway slope and a wet runway enter as

$$\theta = \arctan\left(\frac{\text{slope}_{\%}}{100}\right) \tag{5.1}$$

$$\mu_{\text{brake,wet}} = 0.5\,\mu_{\text{brake}} \tag{5.2}$$

the slope term following [12] and the wet-runway factor a simplification of [11]. The state is tracked in ground speed, while every aerodynamic and thrust force uses true airspeed:

$$V_{\text{TAS}} = V_{\text{GS}} + V_w \tag{5.3}$$

5.2 `accelerate()`

`accelerate(acft, cond, tas_target_mps, gs_start_mps, dist_start_m, engines_operating)` → `(dist_end_m, time_s, solution)`
Returns: distance and time when the target airspeed is reached.

The accelerating ground roll [8, Ch. 6], with $C_L = C_{L,\text{gnd}}$ and the slope term [12]:

$$m\,\frac{dV_{\text{GS}}}{dt} = T(V_{\text{TAS}}) - D(V_{\text{TAS}}) - \mu_{\text{roll}}\,(W - L(V_{\text{TAS}})) - W\sin\theta, \qquad \frac{ds}{dt} = V_{\text{GS}} \tag{5.4}$$

If the aircraft cannot overcome friction at rest, the acceleration is set to zero so it cannot roll backwards. If the target speed is never reached, `InsufficientThrustError` is raised.

5.3 `decelerate()`

`decelerate(acft, cond, gs_start_mps, dist_start_m, cl, use_reverse)` → `(dist_end_m, time_s, solution)`
Returns: distance and time to a stop.

The braking roll at idle thrust (taken as zero) [8, Ch. 6]; [1, Ch. 17]:

$$m\,\frac{dV_{\text{GS}}}{dt} = -\Big[D + \mu_{\text{brake}}\,(W - L) + T_{\text{rev}} + W\sin\theta\Big] \tag{5.5}$$

5.4 Numerical Integration

Both functions integrate the state $\mathbf{y} = [\, s,\ V_{\text{GS}} \,]^{\mathsf{T}}$ with an adaptive Runge–Kutta 4(5) method (SciPy `solve_ivp`, RK45), a maximum step of 0.5 s and a 300 s limit. Each segment ends on an event function crossing zero:

$$g_{\text{accel}}(t, \mathbf{y}) = (V_{\text{GS}} + V_w) - V_{\text{target}} \tag{5.6}$$

$$g_{\text{stop}}(t, \mathbf{y}) = V_{\text{GS}} - 0.1 \tag{5.7}$$

Reducing the step limit from 0.5 s to 0.1 s changes the Takeoff Field Length by less than 0.1 ft.

5.5 `climb_to_screen()`

`climb_to_screen(acft, cond, vs1_mps, screen_height_ft, engines_operating)` → `(air_dist_m, climb_grad_pct)`
Returns: air distance from liftoff to the screen height, and the climb gradient.

Raymer's transition-arc method [1, Ch. 17]. First the climb angle at $V_2 = 1.2 V_{S1}$ with lift equal to weight:

$$C_{L,\text{climb}} = \frac{W}{q_{V_2}\, S} \tag{5.8}$$

$$\sin\gamma = \frac{T(V_2) - D(V_2,\, C_{L,\text{climb}})}{W}, \qquad \text{gradient} = 100\tan\gamma\ [\%] \tag{5.9}$$

If $\sin\gamma \le 0$, `ClimbPerformanceError` is raised. The transition from the ground roll to the climb is a circular arc flown at $V_{\text{TR}} = 1.15 V_{S1}$ with load factor $n = 1.2$:

$$R_{\text{TR}} = \frac{V_{\text{TR}}^2}{g_0\,(n-1)} \tag{5.10}$$

$$h_{\text{TR}} = R_{\text{TR}}\,(1 - \cos\gamma) \tag{5.11}$$

If the screen height is reached inside the arc ($h_{\text{TR}} \ge h_{\text{scr}}$):

$$s_{\text{air}} = \sqrt{R_{\text{TR}}^2 - (R_{\text{TR}} - h_{\text{scr}})^2} \tag{5.12}$$

otherwise the arc is followed by a steady climb:

$$s_{\text{air}} = R_{\text{TR}}\sin\gamma + \frac{h_{\text{scr}} - h_{\text{TR}}}{\tan\gamma} \tag{5.13}$$

Finally the air distance is converted to ground distance for wind:

$$s_{\text{air}} \leftarrow s_{\text{air}}\,\frac{V_{\text{TR}} - V_w}{V_{\text{TR}}} \tag{5.14}$$

6 `takeoff.py`: All-Engines Takeoff Distance

6.1 `takeoff_speeds()`

`takeoff_speeds(acft, cond)` → `(vs1_mps, vr_mps, v2_mps)`
Returns: the takeoff V-speeds (TAS).

$$V_R = 1.1\,V_{S1}, \qquad V_2 = 1.2\,V_{S1}, \qquad V_{\text{TR}} = 1.15\,V_{S1} \tag{6.1}$$

Ratios from [1, Ch. 17], consistent with [6, §25.107.]

6.2 `rotation_distance()`

`rotation_distance(acft, cond, vr_mps)` → `float`
Returns: ground distance while rotating from V_R to liftoff [m].

The aircraft rotates for t_{rot} (1 s light aircraft, 3 s jets [1]) at roughly constant speed:

$$s_{\text{rot}} = (V_R - V_w)\,t_{\text{rot}} \tag{6.2}$$

6.3 `takeoff()`

`takeoff(acft, elev_m, isa_dev_c, mass_kg, hw_kt, slope_pct, rwy_condition)` → `dict`
Returns: V-speeds, ground roll, air distance, climb gradient and TOD.

1. `make_conditions()` builds the day's conditions.
2. `takeoff_speeds()` gives V_{S1}, V_R and V_2.
3. `accelerate()` integrates (5.4) from rest to V_R; `rotation_distance()` adds the rotation:

$$s_{\text{ground}} = s(V_R) + s_{\text{rot}} \tag{6.3}$$

4. `climb_to_screen()` gives the air distance to 50 ft (props) or 35 ft (jets).

$$\text{TOD}_{\text{AEO}} = s_{\text{ground}} + s_{\text{air}} \tag{6.4}$$

7 `bfl.py`: Balanced Field Length and Takeoff Field Length

Method: [13]; [14]; [10]; [11]; rules from [6, §§25.107–25.113.] For a multi-engine aircraft one engine is assumed to fail at the decision speed V_1 (the model takes the engine-failure speed $V_{\text{EF}} = V_1$). For any trial V_1 there are two outcomes.

7.1 `accelerate_go()`

`accelerate_go(acft, cond, v1_mps)` → `(distance_m, oei_climb_grad_pct)`
Returns: distance to 35 ft when the takeoff continues after the failure.

$$s_{\text{go}}(V_1) = \underbrace{s_{\text{AEO}}(0 \to V_1)}_{\text{all engines}} + \underbrace{s_{\text{OEI}}(V_1 \to V_R)}_{\text{one engine out}} + s_{\text{rot}} + s_{\text{air,OEI}} \tag{7.1}$$

Each term uses `accelerate()` or `climb_to_screen()` with the thrust of (4.5).

7.2 `accelerate_stop()`

`accelerate_stop(acft, cond, v1_mps)` → `float`
Returns: distance to a full stop when the takeoff is rejected.

The 2-second allowance at V_1 follows §25.109. Braking uses spoilers ($C_L = 0$), idle thrust and no reverse thrust credit:

$$s_{\text{stop}}(V_1) = s_{\text{AEO}}(0 \to V_1) + (V_1 - V_w)\,(2\,\text{s}) + s_{\text{brake}}(V_1 \to 0) \tag{7.2}$$

7.3 `balanced_field()`

`balanced_field(acft, elev_m, isa_dev_c, mass_kg, hw_kt, slope_pct, rwy_condition)` → `dict`
Returns: balanced V_1, BFL, AEO TOD, TOFL and the curves for the chart.

A low V_1 makes the go distance long and the stop distance short; a high V_1 does the opposite. The balanced V_1 is the root of

$$f(V_1) = s_{\text{go}}(V_1) - s_{\text{stop}}(V_1) = 0, \qquad 0.5\,V_R \le V_1 \le V_R \tag{7.3}$$

found with Brent's method (`scipy.optimize.brentq`, tolerance 0.01 m/s). The upper bound reflects $V_1 \le V_R$ (§25.107). If f does not change sign, V_1 is set to whichever bound applies and the

result is reported as unbalanced. The minimum control speed on the ground (VMCG) is not checked. At the root,

$$\text{BFL} = s_{\text{go}}(V_1) = s_{\text{stop}}(V_1) \tag{7.4}$$

§25.113 requires the takeoff distance to be the greater of the engine-out distance and 115% of the all-engines distance:

$$\text{TOFL} = \max\left(\text{BFL},\ 1.15 \times \text{TOD}_{\text{AEO}}\right) \tag{7.5}$$

7.4 `plot_bfl()`

`plot_bfl(result, filename, runway, extra_conditions)` → `filename`
Returns: the balanced-field chart.

Plots s_{go} and s_{stop} against V_1, marks the balanced point, V_R and the $1.15 \times \text{TOD}_{\text{AEO}}$ line, and optionally the runway's TODA and ASDA.

8 `landing.py`: Landing Distance

8.1 `landing()`

`landing(acft, elev_m, isa_dev_c, mass_kg, hw_kt, use_reverse, slope_pct, rwy_condition)` → `dict`
Returns: landing speeds, air distance, free roll, braking roll, LD and LDR.

Method: [8, Ch. 6]; [1, Ch. 17]; [6, §25.125]; [15, §121.195.] The landing screen height is 50 ft for every aircraft.

$$V_{\text{REF}} = 1.3\,V_{S0}, \qquad V_{\text{TD}} = 1.15\,V_{S0} \tag{8.1}$$

$$s_{\text{air}} = \frac{h_{\text{scr}}}{\tan 3^\circ} \cdot \frac{V_{\text{REF}} - V_w}{V_{\text{REF}}} \tag{8.2}$$

$$s_{\text{free}} = (V_{\text{TD}} - V_w)\,t_{\text{free}}, \qquad t_{\text{free}} = 2\,\text{s (prop)},\ 3\,\text{s (jet)} \tag{8.3}$$

$$\text{LD} = s_{\text{air}} + s_{\text{free}} + s_{\text{brake}} \tag{8.4}$$

where s_{brake} is (5.5) integrated by `decelerate()` from $V_{\text{TD}} - V_w$ to rest. For jets, §121.195 requires the landing to fit within 60% of the runway, with a further 15% on a wet runway:

$$\text{LDR}_{\text{dry}} = \frac{\text{LD}_{\text{dry}}}{0.6}, \qquad \text{LDR}_{\text{wet}} = 1.15 \times \text{LDR}_{\text{dry}} \tag{8.5}$$

9 `airports.py`: Runways and Wind Components

9.1 `Runway` and `Airport`

Each runway direction is stored separately with its true heading, slope and declared distances [16]: Takeoff Run Available (TORA), Takeoff Distance Available (TODA), Accelerate-Stop Distance Available (ASDA) and Landing Distance Available (LDA).

9.2 `runway_pair()`

`runway_pair(id_1, hdg_1, id_2, hdg_2, length_ft, disp_1_ft, disp_2_ft, slope_1_pct)` → `[Runway, Runway]`
Returns: both directions of one physical runway.

Declared distances are approximated from the runway length ℓ and the displaced threshold d at the landing end; the slope comes from the threshold elevations [17]:

$$\text{TORA} = \text{TODA} = \text{ASDA} = \ell, \qquad \text{LDA} = \ell - d \tag{9.1}$$

$$\text{slope}_{\%} = 100\,\frac{h_{\text{far end}} - h_{\text{near end}}}{\ell} \tag{9.2}$$

9.3 `wind_components()`

`wind_components(wind_dir_deg, wind_kt, rwy_hdg_deg)` → `(hw_kt, xw_kt)`
Returns: headwind and crosswind components.

For a wind from ψ_w on a runway of true heading ψ_r [11]:

$$V_w = V_{\text{wind}}\cos(\psi_w - \psi_r) \qquad (+\ \text{headwind}) \tag{9.3}$$

$$V_x = V_{\text{wind}}\sin(\psi_w - \psi_r) \qquad (+\ \text{from the right}) \tag{9.4}$$

10 `runway_analysis.py`: Runway Checks

10.1 `isa_dev_from_oat()`

`isa_dev_from_oat(elev_m, oat_c)` → `float`
Returns: ISA deviation [°C].

$$\Delta T = \text{OAT} - (\Theta_{\text{std}}(h) - 273.15) \tag{10.1}$$

10.2 `factored_headwind()`

`factored_headwind(hw_kt)` → `float`
Returns: the headwind used for performance.

No more than 50% of a headwind may be credited and at least 150% of a tailwind must be applied [6, §25.105(d)](1):

$$V_{w,\text{perf}} = \begin{cases} 0.5\,V_w, & V_w \geq 0 \\ 1.5\,V_w, & V_w < 0 \end{cases} \tag{10.2}$$

10.3 `analyze_runway()`

`analyze_runway(acft, apt, rwy, oat_c, wind_dir_deg, wind_kt, rwy_condition, tow_kg, lw_kg)` → `dict`
Returns: every distance and a pass or fail for takeoff and landing.

The OAT is converted with (10.1), the wind with (9.3)–(10.2), and the runway's slope is applied. The distances are then checked against the declared distances (Table 4).

Aircraft	Takeoff Checks	Landing Check
Multi-engine	TOFL $\leq$ TODA and ASD $\leq$ ASDA	LDR $\leq$ LDA (jets)
Single-engine	TOD $\leq$ TODA and $s_{\text{ground}} \leq$ TORA	LD $\leq$ LDA

Table 4. Runway checks by aircraft type.

11 `sensitivity.py`: Parameter Sweeps

11.1 `takeoff_metric_m()` and `landing_metric_m()`

`takeoff_metric_m(acft, **conditions)`, `landing_metric_m(acft, **conditions)` → `float`
Returns: the distance that matters for the aircraft [m].

The takeoff measure is the TOFL (7.5) for multi-engine aircraft and the AEO TOD (6.4) for single-engine aircraft. The landing measure is the unfactored LD (8.4) with no reverse thrust credit.

11.2 `run_sweeps()`

`run_sweeps(acft)` → `dict`
Returns: the distances over each sweep.

Each variable is changed on its own from the baseline (sea level, ISA, MTOW / MLW, calm wind, dry level runway): field elevation 0 to 8 000 ft, ISA deviation -20 to $+30$ °C, weight 70–100%, headwind -10 to $+20$ kt. The change from the baseline is

$$\Delta_{\%} = 100 \left(\frac{s}{s_{\text{baseline}}} - 1 \right) \tag{11.1}$$

12 Supporting Files

These files contain no performance formulas.

- `exceptions.py`: `PerformanceError` and its subclasses `InsufficientThrustError`, `ClimbPerformanceError` and `AtmosphereRangeError`, so the menu can report exactly why a case cannot be computed.
- `profiles.py`: create, save and load aircraft profiles (JSON), rebuilt as `PropAircraft` or `JetAircraft`.
- `plot_style.py`: the shared chart style.
- `main.py`: the menu that ties everything together.

13 Computation Procedure

For one aircraft, runway and day, the calls run in this order:

1. **Day conditions.** `isa_dev_from_oat()` (10.1); `isa()` (3.1)–(3.4); `wind_components()` and `factored_headwind()` (9.3)–(10.2); `make_conditions()` (5.1)–(5.3).
2. **Speeds.** `stall_speed()` (4.4); `takeoff_speeds()` (6.1).
3. **All-engines takeoff.** `accelerate()` (5.4) to V_R; `rotation_distance()` (6.2); `climb_to_screen()` (5.8)–(5.14); TOD (6.4).
4. **Engine failure (multi-engine only).** `accelerate_go()` (7.1) and `accelerate_stop()` (7.2) for trial V_1; Brent's method on (7.3); BFL (7.4); TOFL (7.5).
5. **Landing.** `landing()`: (8.1)–(8.4), with `decelerate()` (5.5); LDR (8.5).
6. **Runway check.** Compare against TORA, TODA, ASDA and LDA (Table 4).

PART II

Worked Examples

Both examples use the same runway and day, so the two aircraft can be followed side by side through the procedure of Section 13:

- Wichita Dwight D. Eisenhower National (KICT), runway 01L, elevation 1 333 ft
- Outside air temperature 30 °C, wind 20° at 10 kt (straight down the runway)
- Dry runway, slope taken from the threshold elevations
- Takeoff at MTOW, landing at MLW

Every number below is produced by the model itself (`report/make_examples.py`).

14 Example A: Bombardier Challenger 3500

14.1 Inputs

Table 5 lists the aircraft data and the day's conditions. The example uses Wichita Dwight D. Eisenhower National (KICT) runway 01L.

Input	Value
Maximum takeoff weight (MTOW)	40 600 lb = 18 415.9 kg
Maximum landing weight (MLW)	34 150 lb = 15 490.2 kg
Reference wing area, S	48.50 m^2 (522.0 ft^2)
C_{D_0} / k	0.030 / 0.045
$C_{L,\max}$ takeoff / landing	1.57 / 2.30
$C_{L,\text{gnd}}$	0.30
μ_{roll} / μ_{brake}	0.02 / 0.40
Sea-level static thrust, T_{SLS}	65 149 N (14 646 lbf)
Number of engines, N_e	2
Thrust lapse, k_{lapse}	0.90
Flat rating, ΔT_{flat}	ISA 15 degreeCelsius
Reverse thrust ratio	0.30
Field elevation, h	1 333 ft = 406.3 m
Outside air temperature (OAT)	30 degreeCelsius
Reported wind	20° / 10 kt
Runway true heading	20°
Runway slope (direction of travel)	0.165 %
Runway condition	Dry
TODA = ASDA / LDA	10 302 ft / 10 302 ft

Table 5. Challenger 3500: example inputs.

14.2 Step 1: Atmosphere

Standard temperature at the field elevation, then the ISA deviation from the OAT ((3.1) and (10.1)):

$$\Theta_{\text{std}} = 288.15 - 0.0065 \times 406.3 = 285.51\,\text{K}$$
$$\Delta T = 30.0 - (285.51 - 273.15) = 17.64\,\text{degreeCelsius} \quad (\text{ISA } +17.6)$$

Pressure, temperature, density, density ratio and speed of sound ((3.2)–(3.6)):

$$p = 101\,325\left(\frac{285.51}{288.15}\right)^{5.2559} = 96\,538\,\text{Pa}$$
$$\Theta = 285.51 + 17.64 = 303.15\,\text{K}$$
$$\rho = \frac{96\,538}{287.05 \times 303.15} = 1.1094\,\text{kg/m}^3$$
$$\sigma = \frac{1.1094}{1.225} = 0.9056$$
$$a = \sqrt{1.4 \times 287.05 \times 303.15} = 349.0\,\text{m/s}$$

14.3 Step 2: Wind

Runway components ((9.3)–(9.4)), then the regulatory factor (10.2):

$$V_w = 10\cos(20^\circ - 20^\circ) = 10.0\,\text{kt} \qquad V_x = 10\sin(0^\circ) = 0.0\,\text{kt}$$
$$V_{w,\text{perf}} = 0.5 \times 10.0 = 5.0\,\text{kt} = 2.57\,\text{m/s}$$

All following steps use the performance headwind of 5.0 kt.

14.4 Step 3: Weight and Takeoff Speeds

$$W = m\,g_0 = 18\,415.9 \times 9.80665 = 180\,598\,\text{N}$$
$$V_{S1} = \sqrt{\frac{2 \times 180\,598}{1.1094 \times 48.50 \times 1.57}} = 65.39\,\text{m/s} = 127.1\,\text{kt}$$
$$V_R = 1.10\,V_{S1} = 71.93\,\text{m/s} = 139.8\,\text{kt}$$
$$V_2 = 1.20\,V_{S1} = 78.47\,\text{m/s} = 152.5\,\text{kt}$$
$$V_{\text{TR}} = 1.15\,V_{S1} = 75.20\,\text{m/s} = 146.2\,\text{kt}$$

14.5 Step 4: Forces on the Ground Roll

Thrust ((4.8)–(4.10)). The ISA deviation of 17.6 °C exceeds the flat rating by 2.6 °C, so

$$\sigma_{\text{eff}} = \sigma(h,\ 2.6) = 0.9528$$
$$T(V) = 65\,149 \times 0.9528^{0.7}\left(1 - 0.90\,\frac{V}{349.0}\right)$$

Forces at brake release and at V_R ((5.4), $C_L = C_{L,\text{gnd}} = 0.30$, slope force $W \sin\theta$):

Quantity	**At $V_{\text{TAS}} = V_w$ (start)**	**At V_R**
Dynamic pressure q (Pa)	3.7	2 869.8
Lift L (N)	53	41 756
Drag D (N)	6	4 739
Thrust T (N)	62 561	51 298
Rolling friction $\mu_{\text{roll}}(W - L)$ (N)	3 611	2 777
Slope force $W \sin\theta$ (N)	298	298
Acceleration a (m/s^2)	3.185	2.361

Table 6. Challenger 3500: forces on the ground roll.

14.6 Step 5: Ground Roll by Numerical Integration

Equation (5.4) is integrated with RK45 from rest until $V_{\text{TAS}} = V_R$. Selected points from the solver:

t **(s)**	V_{TAS} **(kt)**	a (m/s^2)	s **(ft)**
0.0	5.0	3.185	0
2.6	21.0	3.108	35
7.1	47.5	2.967	258
11.6	72.9	2.819	678
16.1	96.9	2.666	1 286
20.6	119.5	2.511	2 070
24.90	139.8 (V_R)	2.361	2 974

Table 7. Challenger 3500: ground-roll time history (RK45 output).

Rotation for t_{rot} (6.2):

$$s_{\text{rot}} = (71.93 - 2.57) \times 3 = 208.1\,\text{m}$$
$$s_{\text{ground}} = 906.5 + 208.1 = 1\,114.5\,\text{m} = 3\,657\,\text{ft}$$

14.7 Step 6: Air Distance to the Screen Height

Climb at V_2 with lift equal to weight ((5.8)–(5.9)); screen height 35 ft = 10.67 m:

$$q_{V_2} = \tfrac{1}{2} \times 1.1094 \times 78.47^2 = 3\,415.3\ \text{Pa}, \qquad C_{L,\text{climb}} = \frac{180\,598}{3\,415.3 \times 48.50} = 1.090$$

$$D = 3\,415.3 \times 48.50 \times (0.030 + 0.045 \times 1.090^2) = 13\,830\ \text{N}, \qquad T = 50\,237\ \text{N}$$

$$\sin\gamma = \frac{50\,237 - 13\,830}{180\,598} = 0.2016, \qquad \gamma = 0.2030\ \text{rad}, \qquad \text{gradient} = 20.6\,\%$$

Transition arc ((5.10)–(5.11)):

$$R_{\text{TR}} = \frac{75.20^2}{9.807 \times (1.2 - 1)} = 2\,883.1\ \text{m}$$

$$h_{\text{TR}} = 2\,883.1\,(1 - \cos 0.2030) = 59.19\ \text{m}$$

Since $h_{\text{TR}} \geq h_{\text{scr}}$, the screen is cleared inside the arc (5.12):

$$s_{\text{air}} = \sqrt{2\,883.1^2 - (2\,883.1 - 10.67)^2} = 247.8\ \text{m}$$

Corrected for wind (5.14), then the takeoff distance (6.4):

$$s_{\text{air}} = 247.8 \times \frac{75.20 - 2.57}{75.20} = 239.3\ \text{m}$$

$$\text{TOD}_{\text{AEO}} = 1\,114.5 + 239.3 = 1\,353.8\ \text{m} = 4\,442\ \text{ft}$$

14.8 Step 7: Balanced Field Length

For each trial V_1, the accelerate-go and accelerate-stop distances are computed ((7.1)–(7.2)) and their difference $f(V_1)$ (7.3):

V_1 (kt)	s_{go} (ft)	s_{stop} (ft)	$f(V_1)$ (ft)
69.9	7 586	1 298	6 288
87.4	7 094	2 038	5 056
104.9	6 438	2 957	3 481
118.8	5 772	3 828	1 945
129.3	5 176	4 564	612
139.8	4 482	5 375	−893
133.7	4 896	4 896	0.0

Table 8. Challenger 3500: accelerate-go and accelerate-stop against trial V_1.

f changes sign between the trial values, so Brent's method converges to the balanced $V_1 = 133.7$ kt

(bold row). Then ((7.4)–(7.5)):

$$\begin{aligned} \text{BFL} &= 4\,896\,\text{ft} \\ 1.15 \times \text{TOD}_{\text{AEO}} &= 1.15 \times 4\,442 = 5\,108\,\text{ft} \\ \text{TOFL} &= \max\left(4\,896,\ 5\,108\right) = 5\,108\,\text{ft} = 1\,557\,\text{m} \end{aligned}$$

The TOFL is set by the all-engines distance with its 15% margin. The one-engine-inoperative climb gradient at V_2 is 6.3 %.

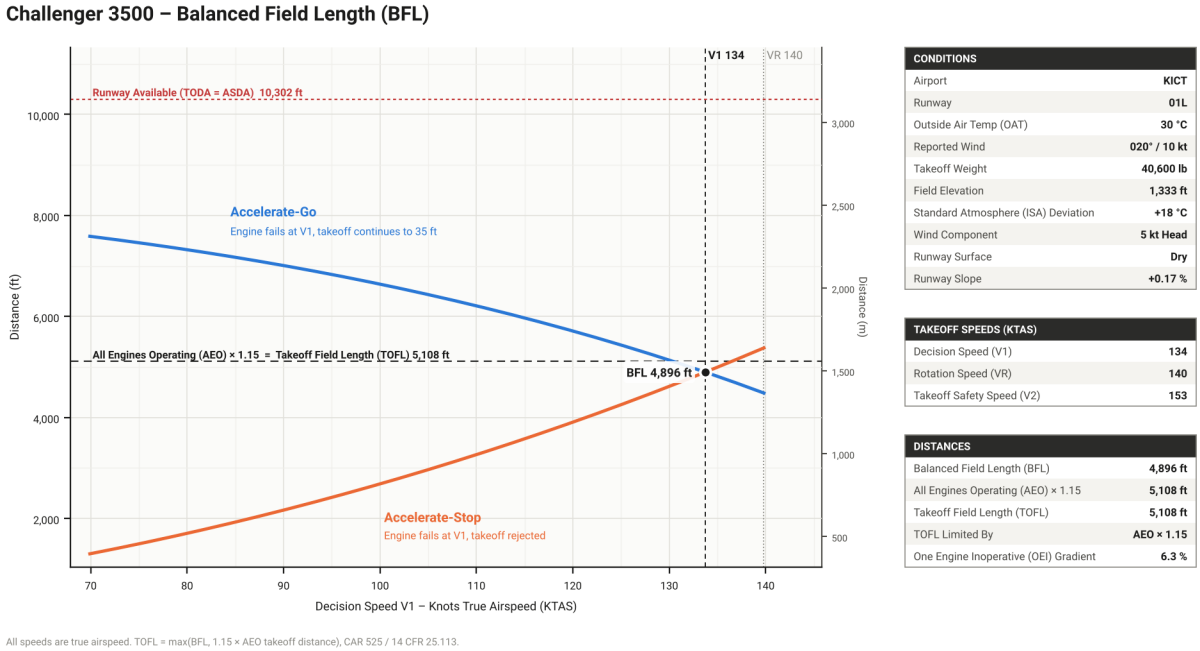


Figure 1. Challenger 3500: Balanced Field Length (BFL) for this example.

14.9 Step 8: Landing

Landing at MLW on the same runway, no reverse thrust credit ((8.1)–(8.5)):

$$
\begin{aligned}
V_{S0} &= \sqrt{\frac{2 \times 151\,907}{1.1094 \times 48.50 \times 2.30}} = 49.55\,\text{m/s} = 96.3\,\text{kt} \\
V_{\text{REF}} &= 1.30\,V_{S0} = 125.2\,\text{kt}, \qquad V_{\text{TD}} = 1.15\,V_{S0} = 110.8\,\text{kt} \\
s_{\text{air}} &= \frac{15.24}{\tan 3^\circ} \times \frac{64.41 - 2.57}{64.41} = 279.2\,\text{m} \\
s_{\text{free}} &= (56.98 - 2.57) \times 3 = 163.2\,\text{m} \\
s_{\text{brake}} &= 367.7\,\text{m} \quad \text{((5.5) integrated to rest, 13.6 s)} \\
\text{LD} &= 279.2 + 163.2 + 367.7 = 810.1\,\text{m} = 2\,658\,\text{ft} \\
\text{LDR} &= \frac{810.1}{0.6} = 1\,350.2\,\text{m} = 4\,430\,\text{ft}
\end{aligned}
$$

14.10 Step 9: Runway Check

Required	ft	Available	ft	Margin (ft)	Result
Takeoff Field Length (TOFL)	5 108	TODA	10 302	5 194	OK
Accelerate-Stop Distance (ASD)	4 896	ASDA	10 302	5 406	OK
Landing Distance Required (LDR)	4 430	LDA	10 302	5 872	OK

Table 9. Challenger 3500: runway check, KICT runway 01L.

15 Example B: Cirrus SR22T

15.1 Inputs

Table 10 lists the aircraft data and the day's conditions. The example uses Wichita Dwight D. Eisenhower National (KICT) runway 01L.

Input	Value
Maximum takeoff weight (MTOW)	3 600 lb = 1 632.9 kg
Maximum landing weight (MLW)	3 600 lb = 1 632.9 kg
Reference wing area, S	13.46 m^2 (144.9 ft^2)
C_{D_0} / k	0.027 / 0.045
$C_{L,\max}$ takeoff / landing	1.60 / 2.04
$C_{L,\text{gnd}}$	0.40
μ_{roll} / μ_{brake}	0.04 / 0.40
Sea-level static thrust, T_{SLS}	3 580 N (805 lbf)
Number of engines, N_e	1
Maximum power, P_{max}	234.9 kW (315 hp)
Propeller efficiency, η_p	0.72
Turbocharger critical altitude	25 000 ft
Field elevation, h	1 333 ft = 406.3 m
Outside air temperature (OAT)	30 degreeCelsius
Reported wind	20° / 10 kt
Runway true heading	20°
Runway slope (direction of travel)	0.165 %
Runway condition	Dry
TODA = ASDA / LDA	10 302 ft / 10 302 ft

Table 10. Cirrus SR22T: example inputs.

15.2 Step 1: Atmosphere

Standard temperature at the field elevation, then the ISA deviation from the OAT ((3.1) and (10.1)):

$$\Theta_{\text{std}} = 288.15 - 0.0065 \times 406.3 = 285.51\,\text{K}$$
$$\Delta T = 30.0 - (285.51 - 273.15) = 17.64\,\text{degreeCelsius} \quad (\text{ISA } +17.6)$$

Pressure, temperature, density, density ratio and speed of sound ((3.2)–(3.6)):

$$p = 101\,325\left(\frac{285.51}{288.15}\right)^{5.2559} = 96\,538\ \text{Pa}$$
$$\Theta = 285.51 + 17.64 = 303.15\ \text{K}$$
$$\rho = \frac{96\,538}{287.05 \times 303.15} = 1.1094\ \text{kg/m}^3$$
$$\sigma = \frac{1.1094}{1.225} = 0.9056$$
$$a = \sqrt{1.4 \times 287.05 \times 303.15} = 349.0\ \text{m/s}$$

15.3 Step 2: Wind

Runway components ((9.3)–(9.4)), then the regulatory factor (10.2):

$$V_w = 10\cos(20^\circ - 20^\circ) = 10.0\ \text{kt} \qquad V_x = 10\sin(0^\circ) = 0.0\ \text{kt}$$
$$V_{w,\text{perf}} = 0.5 \times 10.0 = 5.0\ \text{kt} = 2.57\ \text{m/s}$$

All following steps use the performance headwind of 5.0 kt.

15.4 Step 3: Weight and Takeoff Speeds

$$W = m\,g_0 = 1\,632.9 \times 9.80665 = 16\,014\ \text{N}$$
$$V_{S1} = \sqrt{\frac{2 \times 16\,014}{1.1094 \times 13.46 \times 1.60}} = 36.61\ \text{m/s} = 71.2\ \text{kt}$$
$$V_R = 1.10\,V_{S1} = 40.27\ \text{m/s} = 78.3\ \text{kt}$$
$$V_2 = 1.20\,V_{S1} = 43.93\ \text{m/s} = 85.4\ \text{kt}$$
$$V_{\text{TR}} = 1.15\,V_{S1} = 42.10\ \text{m/s} = 81.8\ \text{kt}$$

15.5 Step 4: Forces on the Ground Roll

Thrust (4.7), turbocharged below its critical altitude so the power is not reduced ($\sigma_P = 1$):

$$T(V) = \min\left(1.0000 \times 3\,580,\ \frac{0.72 \times 1.0000 \times 234\,895}{V}\right)$$

Forces at brake release and at V_R ((5.4), $C_L = C_{L,\text{gnd}} = 0.40$, slope force $W \sin\theta$):

Quantity	**At $V_{\text{TAS}} = V_w$ (start)**	**At V_R**
Dynamic pressure q (Pa)	3.7	899.6
Lift L (N)	20	4 844
Drag D (N)	2	414
Thrust T (N)	3 580	3 580
Rolling friction $\mu_{\text{roll}}(W - L)$ (N)	640	447
Slope force $W \sin\theta$ (N)	26	26
Acceleration a (m/s^2)	1.783	1.649

Table 11. Cirrus SR22T: forces on the ground roll.

15.6 Step 5: Ground Roll by Numerical Integration

Equation (5.4) is integrated with RK45 from rest until $V_{\text{TAS}} = V_R$. Selected points from the solver:

t **(s)**	V_{TAS} **(kt)**	a **(m/s^2)**	s **(ft)**
0.0	5.0	1.783	0
2.1	12.3	1.781	13
6.1	26.1	1.769	109
10.1	39.8	1.749	298
14.1	53.3	1.721	579
18.1	66.5	1.686	949
21.73	78.3 (V_R)	1.649	1 361

Table 12. Cirrus SR22T: ground-roll time history (RK45 output).

Rotation for t_{rot} (6.2):

$$s_{\text{rot}} = (40.27 - 2.57) \times 1 = 37.7\,\text{m}$$

$$s_{\text{ground}} = 414.9 + 37.7 = 452.6\,\text{m} = 1\,485\,\text{ft}$$

15.7 Step 6: Air Distance to the Screen Height

Climb at V_2 with lift equal to weight ((5.8)–(5.9)); screen height 50 ft = 15.24 m:

$$q_{V_2} = \tfrac{1}{2} \times 1.1094 \times 43.93^2 = 1\,070.6\,\text{Pa}, \qquad C_{L,\text{climb}} = \frac{16\,014}{1\,070.6 \times 13.46} = 1.111$$

$$D = 1\,070.6 \times 13.46 \times (0.027 + 0.045 \times 1.111^2) = 1\,190\,\text{N}, \qquad T = 3\,580\,\text{N}$$

$$\sin\gamma = \frac{3\,580 - 1\,190}{16\,014} = 0.1493, \qquad \gamma = 0.1498\,\text{rad}, \qquad \text{gradient} = 15.1\,\%$$

Transition arc ((5.10)–(5.11)):

$$R_{\text{TR}} = \frac{42.10^2}{9.807 \times (1.2 - 1)} = 903.8\,\text{m}$$

$$h_{\text{TR}} = 903.8\,(1 - \cos 0.1498) = 10.12\,\text{m}$$

Since $h_{\text{TR}} < h_{\text{scr}}$, the arc is followed by a steady climb (5.13):

$$s_{\text{air}} = 903.8 \sin 0.1498 + \frac{15.24 - 10.12}{\tan 0.1498} = 134.9 + 33.9 = 168.8\,\text{m}$$

Corrected for wind (5.14), then the takeoff distance (6.4):

$$s_{\text{air}} = 168.8 \times \frac{42.10 - 2.57}{42.10} = 158.5\,\text{m}$$

$$\text{TOD}_{\text{AEO}} = 452.6 + 158.5 = 611.1\,\text{m} = 2\,005\,\text{ft}$$

15.8 Step 7: Engine-Failure Case

The aircraft has one engine, so there is no accelerate-go case and no balanced field length. The required takeoff distance is the all-engines TOD.

15.9 Step 8: Landing

Landing at MLW on the same runway, no reverse thrust credit ((8.1)–(8.5)):

$$
\begin{aligned}
V_{S0} &= \sqrt{\frac{2 \times 16\,014}{1.1094 \times 13.46 \times 2.04}} = 32.42\,\text{m/s} = 63.0\,\text{kt} \\
V_{\text{REF}} &= 1.30\,V_{S0} = 81.9\,\text{kt}, \qquad V_{\text{TD}} = 1.15\,V_{S0} = 72.5\,\text{kt} \\
s_{\text{air}} &= \frac{15.24}{\tan 3^\circ} \times \frac{42.15 - 2.57}{42.15} = 273.1\,\text{m} \\
s_{\text{free}} &= (37.29 - 2.57) \times 2 = 69.4\,\text{m} \\
s_{\text{brake}} &= 171.9\,\text{m} \quad \text{((5.5) integrated to rest, 9.5 s)} \\
\text{LD} &= 273.1 + 69.4 + 171.9 = 514.4\,\text{m} = 1\,688\,\text{ft}
\end{aligned}
$$

15.10 Step 9: Runway Check

Required	ft	Available	ft	Margin (ft)	Result
Takeoff Distance (TOD)	2 005	TODA	10 302	8 297	OK
Takeoff ground roll	1 485	TORA	10 302	8 817	OK
Landing Distance (LD)	1 688	LDA	10 302	8 614	OK

Table 13. Cirrus SR22T: runway check, KICT runway 01L.

16 Assumptions and Limitations

- Engine failure at $V_{\text{EF}} = V_1$, with no recognition delay before V_1; VMCG not checked.
- Fixed V-speed ratios ($V_R = 1.1V_{S1}$, $V_2 = 1.2V_{S1}$); no separate minimum-unstick speed (VMU) or liftoff speed (VLOF) schedule.
- Wet runway modelled only as halved braking friction; the Part 25 wet-runway takeoff rules are not modelled.
- OEI gradient reported gear-down at V_2; second-segment climb and weight, altitude and temperature (WAT) limits not computed.
- Idle thrust taken as zero when braking; brake energy limits ignored.
- Field elevation treated as pressure altitude; turbocharged power taken as independent of temperature below the critical altitude.
- Declared distances approximated from runway length and displaced thresholds.
- Aircraft aerodynamic coefficients are estimates. For each aircraft one value is calibrated to published data: $C_{L,\max}$ with takeoff flaps for the Challenger 3500 (to its 4 835 ft takeoff field

length [18]) and the static thrust for the SR22T (to its 1 517 ft ground roll [19]).

References

[1] D. P. Raymer, *Aircraft Design: A Conceptual Approach*, 6th ed. Reston, VA, USA: AIAA, 2018.

[2] A. Thompson and B. N. Taylor, "Guide for the use of the International System of Units (SI)," NIST, Gaithersburg, MD, USA, Spec. Publ. 811, 2008.

[3] J. D. Anderson, *Introduction to Flight*, 8th ed. New York, NY, USA: McGraw-Hill, 2016.

[4] International Civil Aviation Organization, *Manual of the ICAO Standard Atmosphere*, Doc 7488/3, 3rd ed. Montréal, QC, Canada: ICAO, 1993.

[5] *Airworthiness Standards: Normal Category Airplanes*, 14 C.F.R. Part 23.

[6] *Airworthiness Standards: Transport Category Airplanes*, 14 C.F.R. Part 25.

[7] Transport Canada, *Canadian Aviation Regulations*, Airworthiness Manual Ch. 525, *Transport Category Aeroplanes*.

[8] J. D. Anderson, *Aircraft Performance and Design*. New York, NY, USA: McGraw-Hill, 1999.

[9] S. Gudmundsson, *General Aviation Aircraft Design: Applied Methods and Procedures*. Oxford, U.K.: Butterworth-Heinemann, 2014.

[10] A. Filippone, *Advanced Aircraft Flight Performance*. Cambridge, U.K.: Cambridge Univ. Press, 2012.

[11] Airbus, "Getting to grips with aircraft performance," Flight Operations Support & Line Assistance, Blagnac, France, 2002.

[12] S. K. Ojha, *Flight Performance of Aircraft*. Washington, DC, USA: AIAA, 1995.

[13] E. Torenbeek, *Synthesis of Subsonic Airplane Design*. Delft, The Netherlands: Delft Univ. Press, 1982.

[14] W. A. Mair and D. L. Birdsall, *Aircraft Performance*. Cambridge, U.K.: Cambridge Univ. Press, 1992.

[15] *Landing Limitations: Destination Airports*, 14 C.F.R. §121.195.

[16] International Civil Aviation Organization, *Annex 14 to the Convention on International Civil Aviation: Aerodromes*, vol. I. Montréal, QC, Canada: ICAO.

[17] OurAirports, "Runway data." [Online]. Available: https://ourairports.com. Accessed: Oct. 21, 2026.

[18] Bombardier, “Challenger 3500 factsheet,” Montréal, QC, Canada, 2024.

[19] Aircraft Owners and Pilots Association, “Aircraft guide: Bombardier Challenger 350; Cirrus SR22T.” [Online]. Available: https://www.aopa.org

A Model Constants

Constant	Value	Meaning	Source
V_R/V_{S1}	1.1	Rotation speed ratio	[1]
V_2/V_{S1}	1.2	Takeoff safety speed ratio	[1], [6]
V_{TR}/V_{S1}	1.15	Transition speed ratio	[1]
n	1.2	Load factor in the transition arc	[1]
t_{rot}	1 s / 3 s	Rotation time, prop / jet	[1]
h_{scr}, takeoff	50 ft / 35 ft	Screen height, Part 23 / Part 25	[5], [6]
h_{scr}, landing	50 ft	Screen height, all aircraft	[6]
t_{stop}	2 s	Allowance at V1 before stopping	[6] §25.109
AEO factor	1.15	$\text{TOFL} \geq 1.15\,\text{TOD}_{\text{AEO}}$	[6] §25.113
V_{REF}/V_{S0}	1.3	Landing reference speed ratio	[1]
V_{TD}/V_{S0}	1.15	Touchdown speed ratio	[1]
Glideslope	3°	Approach path angle	[11]
t_{free}	2 s / 3 s	Free roll, prop / jet	[1]
Reverser cutoff	60 kt	Reversers stowed below	[11]
LDR factor	$1/0.6$	Landing within 60% of runway	[15] §121.195
Wet LDR factor	1.15	Wet runway	[15] §121.195
Wind factors	0.5 / 1.5	Headwind / tailwind	[6] §25.105
Wet braking	$0.5\,\mu_{\text{brake}}$	Wet runway friction	simplified
Integration step	0.5 s	RK45 maximum step	–

Table 14. Constants used in the model.